

# Experimental Setup

Can upstream characteristic records predict held-out local clock reception, including local interaction markers and pulse overlap?

Experiment: gross.reception.v1      Model: signal-space.ss-ocf-1.flat-neutral-clock.v1

Runtime configuration: fixtures/research/gross-test-07.json

Plan lock: 1bd9fb8ce1db34fbbbe0ba2bae4d4ca3ce9455869b9822b050bfd622adbe8f154

## MODEL AND NUMERICAL SETUP

Action: SS OCF 1 flat action,  $U=s-s^2+s^3$ ,  $V=0.25-0.4s+0.2s^2$ ,  $\zeta=0.1$ . Neutral response extension  $Z=1+0.2s$  ( $M_0=m=1$ ); same underlying action, separate registered active-neutral sector.

Units:  $c=\hbar=m=1$ ;  $r,t$  in  $m^{-1}$ ; fields in  $m$ ; frequencies in  $m$ ; dimensionless tick shift

$\Omega\omega\delta t/(2\pi)$ .

Domain:  $R=80$  at  $h=.1,.05,.025$ ;  $R=160$  at  $h=.1$ . Frozen profiles/modes;  $h=.025$  interpolates frozen .05 calibration. Local probe  $r=.1$ ; upstream  $R_s=14$ .

Boundaries: Regular origin and zero radial fields at finite  $R$ . Longevity uses original logged sponge beyond  $r=60$ . Response uses no sponge; earliest outer reflection cannot reach core by  $t=100$ .

Initial Data: Frozen  $\omega_Q=.9$  and  $\Omega\omega=.41274991$ , clock peak .001, phases 0 and  $\pi/3$ . Two compact  $b=r*a$  pulses centered 18 and 26, width3, coefficient  $A=.002,.004,.008$  and negative nominal. Both start exterior to the upstream sphere. First reflected pulse meets second inward pulse around  $r=4$ . Regular-origin reflection, not an independently prepared inner shell or two distant beams.

Solver: Unchanged discrete reciprocal Hamiltonian and RK4;  $dt=.02,.01,.005$  plus independent .01 at  $h=.1$ . Surface acquisition: first-order neutral field on frozen core, records  $a,a_t,a_r$  at  $R_s$  through  $t_{55}$ , no interior histories. Reconstruct  $I=(R_s*a_t+a*R_s*a_r)/2=F_{\text{prime}}$  on  $t_{0..16}$ , integrate cubic spline with  $F(0)=0$  and linear endpoint-closure correction, set  $b(r,0)=F(r-R_s)$ ,  $b_t=F_{\text{prime}}$ . Vacuum exterior is an explicit approximation. Evolve full time-dependent Taylor response on matched evolving quiet background; no adiabatic approximation, no fit. Save all predictions before ANY nonlinear receiver evolution. Sample local fields/derivatives .1, spatial fields every1.

Tolerances: Direct difference error budgets; no Richardson extrapolation. Cycle floor1e-11, field floor1e-13. Local markers first rising and last falling  $|a|=1e-4$  in local acquisition  $t_{0..100}$ . Local phase  $\theta=\text{unwrap atan2}(-\dot{\chi}/\Omega\omega,\chi)$ .  $\delta_N$  is pulse minus quiet phase accumulated between those events divided by  $2\pi$ . Quiet evaluated at the same marker proper times is explicitly counterfactual, no quiet reception claimed. Forecast both marker times and phase before nonlinear evolution; compare standalone interval forecasts. Frozen-core response is also executed. See locked inequalities.

## COMPETING PREDICTIONS

usable: The frozen calibration survives full refined longevity and predicts an even, amplitude-squared local response within registered error controls.

inadequate: Unstable clock, sign-odd response, persistent prediction residual or ambiguous local ticks invalidates the tested approximation.

# Experimental Setup

## CONTROLS

frozen: Prevent calibration retuning and interior information leakage; method: Frozen hashes; predictor consumes only surface records and quiet calibration; prediction locks precede receiver-start events.  
numerics: Bound grid, integration, boundary and small-record errors; method:  $h=.1, .05, .025$ ; independent  $dt/2$  and  $R^2$ ; direct differences plus absolute noise floors.  
pulse-controls: Separate quadratic forcing, overlap and clock phase dependence; method: Quiet, each pulse, both pulses,  $\pm A$ ,  $A/2, A, 2A$ ;  $\pi/3$  clock phase and its quiet baseline; executed frozen-core replay.

## PREREGISTERED DECISIONS

### frozen-inputs (frozen-inputs)

Pass: Saved Test 6 profile/eigenmode hashes match exactly; finer  $h=.025$  uses documented natural cubic interpolation of the frozen  $h=.05$  data, no calibration fit.  
Fail: Resolved finite evidence violates the preregistered inequality.  
Unresolved: Required signal or markers missing or below five-error resolution requirement; no scientific promotion.  
Uncertainty: Direct finer/fine + base/time + base/wide differences and the locked absolute noise floors.

### prediction-first (prediction-first)

Pass: Surface, reconstructed inputs, calibration and all prediction hashes verify; all prediction locks precede the first nonlinear receiver-start event.  
Fail: Resolved finite evidence violates the preregistered inequality.  
Unresolved: Required signal or markers missing or below five-error resolution requirement; no scientific promotion.  
Uncertainty: Direct finer/fine + base/time + base/wide differences and the locked absolute noise floors.

### record-resolution (record-resolution)

Pass: For nominal single, second, both and changed-phase interval records on  $h=.025$ ,  $|\delta_N| > 5*(|finer-fine|+|base-time|+|base-wide|+1e-11 \text{ cycles})$ . Otherwise unresolved.  
Fail: Resolved finite evidence violates the preregistered inequality.  
Unresolved: Required signal or markers missing or below five-error resolution requirement; no scientific promotion.  
Uncertainty: Direct finer/fine + base/time + base/wide differences and the locked absolute noise floors.

### surface-prediction (surface-prediction)

Pass: Fine-grid ( $h=.025$ ) local chi trace prediction relative L2 error  $\leq \max(5\%, \text{direct mesh+time+box L2 error plus } 1e-13 \text{ m/sample floor, divided by signal norm})$ , for single, second, both and phase controls.  
Fail: Resolved finite evidence violates the preregistered inequality.  
Unresolved: Required signal or markers missing or below five-error resolution requirement; no scientific promotion.  
Uncertainty: Direct finer/fine + base/time + base/wide differences and the locked absolute noise floors.

# Experimental Setup

## marker-prediction (marker-prediction)

Pass: For each selected local interval, standalone precomputed marker/phase record differs by  $\leq \max(5\%$  of actual interval record, its registered absolute error budget). If interval not resolved, unresolved.  
Fail: Resolved finite evidence violates the preregistered inequality.  
Unresolved: Required signal or markers missing or below five-error resolution requirement; no scientific promotion.  
Uncertainty: Direct finer/fine + base/time + base/wide differences and the locked absolute noise floors.

## local-markers (local-markers)

Pass: Every nonquiet case on every grid has a first rising and subsequent last falling  $|a|=1e-4$  local crossing in  $t=0..100$ ; predicted marker times within  $0.1 \text{ m}^{-1}$  of observed events. Quiet has no neutral markers and is only a matched counterfactual reference.  
Fail: Resolved finite evidence violates the preregistered inequality.  
Unresolved: Required signal or markers missing or below five-error resolution requirement; no scientific promotion.  
Uncertainty: Direct finer/fine + base/time + base/wide differences and the locked absolute noise floors.

## sign-even (sign-even)

Pass: Single and pair sign-odd/even local trace norm ratio  $<1e-5$  on every grid.  
Fail: Resolved finite evidence violates the preregistered inequality.  
Unresolved: Required signal or markers missing or below five-error resolution requirement; no scientific promotion.  
Uncertainty: Direct finer/fine + base/time + base/wide differences and the locked absolute noise floors.

## amplitude-square (amplitude-square)

Pass:  $A=.002, .004, .008$  amplitude-normalized single-pulse trace differences  $<5\%$ ; weak log slope within  $0.1$  of two on every grid.  
Fail: Resolved finite evidence violates the preregistered inequality.  
Unresolved: Required signal or markers missing or below five-error resolution requirement; no scientific promotion.  
Uncertainty: Direct finer/fine + base/time + base/wide differences and the locked absolute noise floors.

## pulse-overlap (pulse-overlap)

Pass: Both-minus-each-alone chi trace is  $>5$  times direct mesh+time+box+ $1e-13$  m/sample budget and its predicted residual  $\leq \max(5\%, \text{inverse signal/error ratio})$ ; otherwise unresolved.  
Fail: Resolved finite evidence violates the preregistered inequality.  
Unresolved: Required signal or markers missing or below five-error resolution requirement; no scientific promotion.  
Uncertainty: Direct finer/fine + base/time + base/wide differences and the locked absolute noise floors.

# Experimental Setup

frozen-core-control (frozen-core-control)

Pass: Executed incomplete Taylor replay gives max absolute  $\Delta\chi^2$  and  $\Delta\chi^2_{\text{dot2}} < 1e-15$ , while full single response RMS  $> 5e-13$  m.

Fail: Resolved finite evidence violates the preregistered inequality.

Unresolved: Required signal or markers missing or below five-error resolution requirement; no scientific promotion.

Uncertainty: Direct finer/fine + base/time + base/wide differences and the locked absolute noise floors.

numerical-controls (numerical-controls)

Pass: Each relative  $\chi$  response L2 difference (fine/base, time/base, wide/base, finer/fine)  $< 5\%$  in single, second, both and phase controls.

Fail: Resolved finite evidence violates the preregistered inequality.

Unresolved: Required signal or markers missing or below five-error resolution requirement; no scientific promotion.

Uncertainty: Direct finer/fine + base/time + base/wide differences and the locked absolute noise floors.

conservation (conservation)

Pass: Maximum charge relative error  $< 1e-5$  and matched discrete energy residual/initial incident energy  $< 1\%$  for every grid and nonquiet case.

Fail: Resolved finite evidence violates the preregistered inequality.

Unresolved: Required signal or markers missing or below five-error resolution requirement; no scientific promotion.

Uncertainty: Direct finer/fine + base/time + base/wide differences and the locked absolute noise floors.

## VISUAL QUESTIONS

surface-characteristics [evidentiary]: Can the upstream record distinguish incoming drive from returning radiation?

View: Question-driven PNG/SVG/PDF with exact CSV/JSON; observables: local fields, local phase record, numerical controls

usable: resolved predicted even quadratic response

inadequate: prediction mismatch or unresolved local interval

Controls: frozen, numerics, pulse-controls; uncertainty: Grid/time/box differences; radial restriction; counterfactual quiet reference

Data: plot-data/surface.csv; transforms: Saved samples, matched subtraction, prescribed amplitude normalization and local atan2 phase; no fitted response.

packet-overlap [evidentiary]: Where do the reflected first pulse and incoming second pulse overlap?

View: Question-driven PNG/SVG/PDF with exact CSV/JSON; observables: local fields, local phase record, numerical controls

# Experimental Setup

usable: resolved predicted even quadratic response

inadequate: prediction mismatch or unresolved local interval

Controls: frozen, numerics, pulse-controls; uncertainty: Grid/time/box differences; radial restriction; counterfactual quiet reference

Data: plot-data/spacetime.csv; transforms: Saved samples, matched subtraction, prescribed amplitude normalization and local atan2 phase; no fitted response.

withheld-response [evidentiary]: Does surface-only prediction match the withheld local clock and density response?

View: Question-driven PNG/SVG/PDF with exact CSV/JSON; observables: local fields, local phase record, numerical controls

usable: resolved predicted even quadratic response

inadequate: prediction mismatch or unresolved local interval

Controls: frozen, numerics, pulse-controls; uncertainty: Grid/time/box differences; radial restriction; counterfactual quiet reference

Data: plot-data/traces.csv, plot-data/summary.json; transforms: Saved samples, matched subtraction, prescribed amplitude normalization and local atan2 phase; no fitted response.

local-markers [evidentiary]: Do locally triggered intervals carry a resolved predicted clock record?

View: Question-driven PNG/SVG/PDF with exact CSV/JSON; observables: local fields, local phase record, numerical controls

usable: resolved predicted even quadratic response

inadequate: prediction mismatch or unresolved local interval

Controls: frozen, numerics, pulse-controls; uncertainty: Grid/time/box differences; radial restriction; counterfactual quiet reference

Data: plot-data/markers.csv, plot-data/traces.csv, plot-data/summary.json; transforms: Saved samples, matched subtraction, prescribed amplitude normalization and local atan2 phase; no fitted response.

controls [evidentiary]: Do sign, amplitude, pulse-overlap and clock-phase controls support the response mechanism?

View: Question-driven PNG/SVG/PDF with exact CSV/JSON; observables: local fields, local phase record, numerical controls

usable: resolved predicted even quadratic response

inadequate: prediction mismatch or unresolved local interval

Controls: frozen, numerics, pulse-controls; uncertainty: Grid/time/box differences; radial restriction; counterfactual quiet reference

Data: plot-data/traces.csv, plot-data/summary.json; transforms: Saved samples, matched subtraction, prescribed amplitude normalization and local atan2 phase; no fitted response.

error-budget [evidentiary]: Is the local record larger than independent numerical errors?

# Experimental Setup

View: Question-driven PNG/SVG/PDF with exact CSV/JSON; observables: local fields, local phase record, numerical controls  
usable: resolved predicted even quadratic response  
inadequate: prediction mismatch or unresolved local interval  
Controls: frozen, numerics, pulse-controls; uncertainty: Grid/time/box differences; radial restriction; counterfactual quiet reference  
Data: plot-data/summary.json; transforms: Saved samples, matched subtraction, prescribed amplitude normalization and local atan2 phase; no fitted response.

## RESOURCE LIMITS

```
{"max_memory_mb": 1024, "max_output_mb": 250, "max_wall_seconds": 3600}
```

## PROHIBITED INFERENCES

No emergent spacetime, gravity, protected knots or angular stability claim.  
Central reflection supplies counterpropagating radial shells; no two-object exchange or recoil.  
Quiet marker reference is counterfactual; transformed-coordinate invariance and autonomous detector hardware are not evaluated.  
A failed or unresolved marker gate does not become a pass because a full-history trace matches.